

SETU

सेतु · the bridge

An offline comprehension engine for India's language gap.

It does not translate a conversation. It checks that the person understood it — entirely on the laptop, with the network switched off.

Snapdragon® AI Lab Build & Present Challenge

The problem nobody measures

A patient is told, in English, that they have hypertension. **Take amlodipine 5 mg od. Get a fasting lipid profile. Come back immediately if you get chest pain.**

They nod. They say “haan ji.” They go home.

They did not understand “od”. They did not understand “fasting”. And they do not know that chest pain means return now, not at the follow-up.

This is not a translation problem — a phone can translate. It is a comprehension problem, and its defining feature is that nobody in the room ever finds out.

~50%

adherence to long-term therapy
in developing countries (WHO)

1 billion+

Indians who do not speak the language
their own paperwork is written in

What SETU does

A laptop on the desk listens to the whole visit. Wi-Fi off.

1 Listen

Voice activity gates the mic so only real speech reaches Whisper. Continuous, all day, on battery.

2 Bridge

Every turn transcribed, language-identified, shown in the patient's own language.

3 Flag jargon

“od”, “fasting”, “lipid profile” — caught as spoken, glossed in plain words.

4 Build the plan

Instructions extracted as they are given: medicines, tests, follow-up, red flags.

5 Teach back

The patient repeats the plan. SETU checks it against what was actually said.

The moment that matters

The doctor asks the patient to say the plan back. SETU scores it.

2 of 5 understood

The patient did not repeat these back — and they are the ones that matter.

X “If you get chest pain or breathlessness, come immediately to the emergency.”

The doctor learns this while the patient is still in the room — and the patient leaves with a printed card, in their own language, ordered by what will hurt them if they forget it.

Anyone can build a translator. This is a clinical safety instrument.

One engine, three settings

Strip the vocabulary away and these are the same problem: two people, a knowledge gap, one side issuing instructions — and nobody checking they landed.

Clinic

doctor → patient

Catches

od · fasting · lipid profile

Card ordered by

Red flags first

Classroom

teacher → student

Catches

weightage · internal · plagiarism

Card ordered by

Exam dates first

Counter

officer → citizen

Catches

self attested · BPL · acknowledgement

Card ordered by

Deadline first

Adding a courtroom or a bank means writing one Domain object — a lexicon, categories, questions, headings. Not another app. The test suite fails if one domain's vocabulary leaks into another.

Why this needs Snapdragon

Only the third of these is a performance argument.

Un-cloudable by law

A consultation is protected health information. “Send the audio to a data centre” is not a design choice we get to make. On-device is the requirement, not the optimisation.

Works where the link does not

A primary health centre with an intermittent connection is the normal case, not the edge case.

Sustained, not bursty

Continuous speech recognition across a six-hour clinic day is precisely what an NPU exists for and what a CPU cannot afford. Measurable — so we measure it.

Hexa-Router

Most on-device demos hard-code “put everything on the NPU”. That is wrong for a real workload — the NPU is shared and finite, and the right answer changes the moment the charger comes out.

Deployment envelope

What each model can run on at all

Live power state

AC vs battery, charge level, thermal pressure

Request priority

Interactive · streaming · background

Learned cost model

Measured latency and energy, on THIS machine

Decides placement per model, per request — NPU / GPU / CPU

Hysteresis

Rebuilding a QNN session costs hundreds of ms, so a candidate must beat the incumbent by a margin before it is worth paying for.

Thermal backstop

Work is steered off a throttling NPU before it costs more than it saves.

HTP performance mode

burst for interactive turns, sustained_high_performance for all-day streaming, power_saver under battery pressure.

Does it actually work?

Most submissions say “we built it”. We measured it, on a labelled corpus of sessions across all three settings.

0

dangerous misses

Told the doctor the patient understood
when they had not

100%

recall on missed

Of instructions truly not restated, how
many we caught

86.7%

precision

Of what we flagged, how much was
genuinely missed

99

tests passing

Runs green on a bare checkout with no
models present

The evaluation overturned our own design

Replacing keyword matching with sentence embeddings made the check WORSE — precision fell 81% → 76%. Cosine scores an imperative against a first-person promise lower than expected, and the covered/missed ranges overlap. The two signals failed on different items, so the shipped scorer unions them: 86.7%, better than either alone.

Thresholds were chosen at the operating point that keeps dangerous misses at zero — not the one that maximises F1. A false alarm costs a doctor five seconds. A dangerous miss costs a patient.

What runs today — and what does not

Real inference, verified

- Whisper speech recognition — verified on real audio
- Live microphone → transcript → care plan
- Translation into 8 Indian languages
- Sentence embeddings, voice activity detection
- Hexa-Router with measured battery draw

Honestly not finished

- No Snapdragon device — AI Hub profiling harness is built and one command away
- Reasoning path still uses the grounded extractive backend, not the 3B LLM
- Marathi, Bengali, Punjabi have no working checkpoint — the UI says so
- Document capture (OCR) is stubbed

Every fallback is labelled

The whole product runs on a reviewer's laptop with zero models downloaded. Anything on a fallback path is marked degraded in the API and on screen. SETU never passes a stub off as a real inference — in a clinical tool, that distinction is the difference between a demo and a lie.

Not a translator. A check that the patient understood.

Offline, because it has to be.

9

models orchestrated

3

settings, one engine

12

Indian languages

99

tests passing

Full source, architecture notes, benchmarks and evaluation in the repository.